

Deep-sea like OM: Data & Multi-area Projections (Lophius, WKLIFE)

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Introduction

This report builds a deep-sea-like operating model (OM) based on the life-history parameters of *Lophius* (monkfish), as used in WKLIFE. It combines two stages that previously lived in separate scripts, run here in a single knit:

1. **Data build:** derive growth, weight, natural mortality, maturity, selectivity, and a stock-recruitment relationship from life-history parameters, then build an FLBRP/FLStock (`stk0`) and apply a historical fishing trajectory at `Fmsy`.
2. **Operating model:** split `stk0` into three areas — two submarine mounts subject to fishing (`a` and `b`) and one refuge area not subject to fishing (`c`) — sharing a common stock-recruitment relationship, and project forward under three movement scenarios.

Because everything runs in one document, the objects produced in Part 1 (`stk0`, `sr`, `a50`, ...) are simply reused in Part 2 — no intermediate `.RData` file is written or loaded.

```
install.packages(c("tidyr", "dplyr", "ggplot2"))
install.packages(pkgs = c("FLCore", "FLasher", "FLife"),
  repos = c(FLR = "https://flr.r-universe.dev",
    CRAN = "https://cloud.r-project.org"))
```

```
library(FLife)
library(FLasher)
library(ggplot2)
library(tidyr)
library(dplyr)
```

Part 1: Generate a stock from life-history parameters (FLife)

Life-history parameters

```
# stocks and life-history parameters
stocks_lh <- read.csv("stocks.csv")

# select lophius
lh <- stocks_lh[1, ]
agevec <- seq(0, 90, 2)
```

Growth

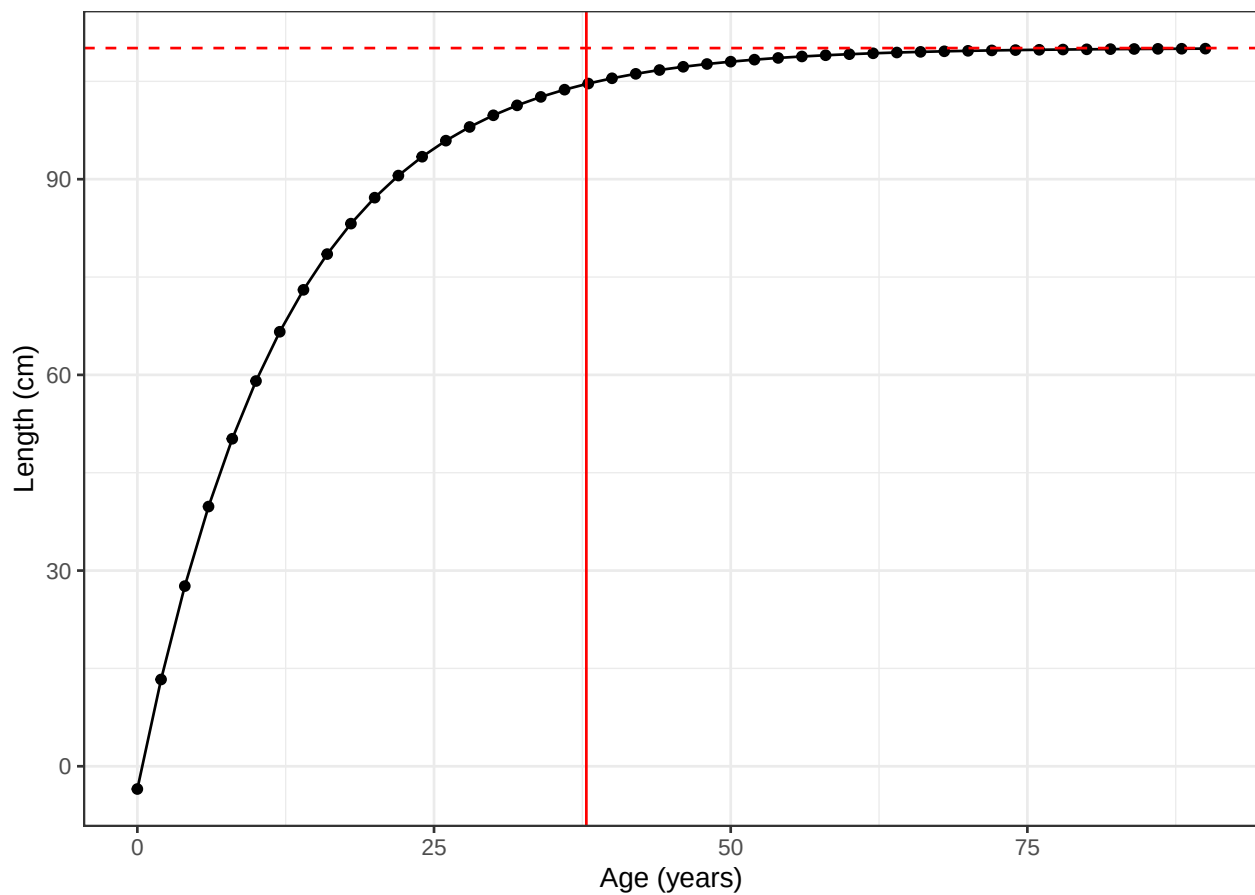
```
# von Bertalanffy parameters  
lh[c("linf", "k", "t0")]
```

```
##      linf      k      t0  
## 1 110.1 0.08 0.39
```

```
tmax <- lh$t0 - log(0.05) / lh$k
```

```
df_length <- data.frame(age = agevec)  
df_length$length <- vonB(df_length$age, params = FLPar(lh))
```

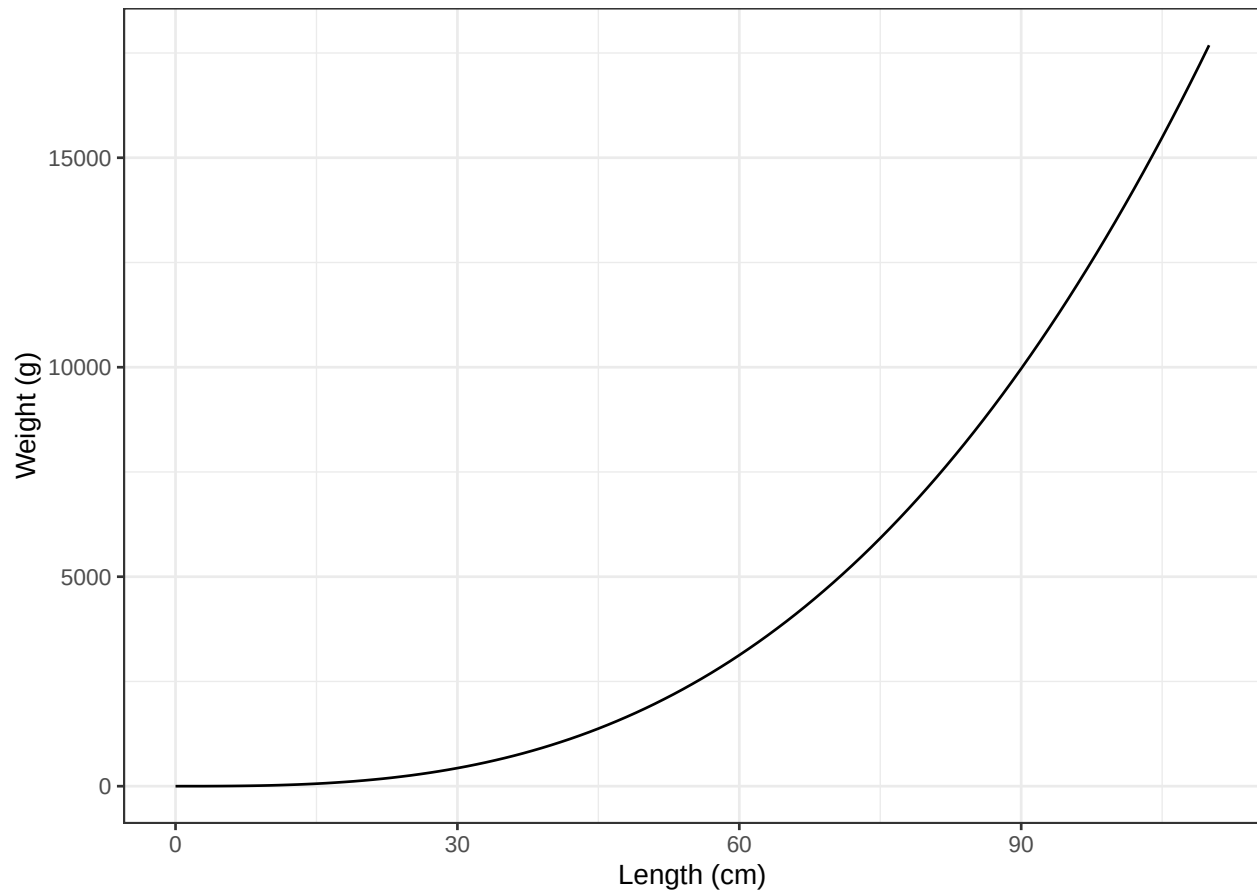
```
df_length %>%  
  ggplot(aes(x = age, y = length)) +  
  geom_line() +  
  geom_point(data = . %>% filter(age %in% agevec)) +  
  geom_hline(yintercept = lh$linf, colour = "red", linetype = "dashed") +  
  geom_vline(xintercept = tmax, colour = "red") +  
  labs(x = "Age (years)", y = "Length (cm)") +  
  theme_bw()
```



Length-weight relationship

```
lh[c("a", "b")]
```

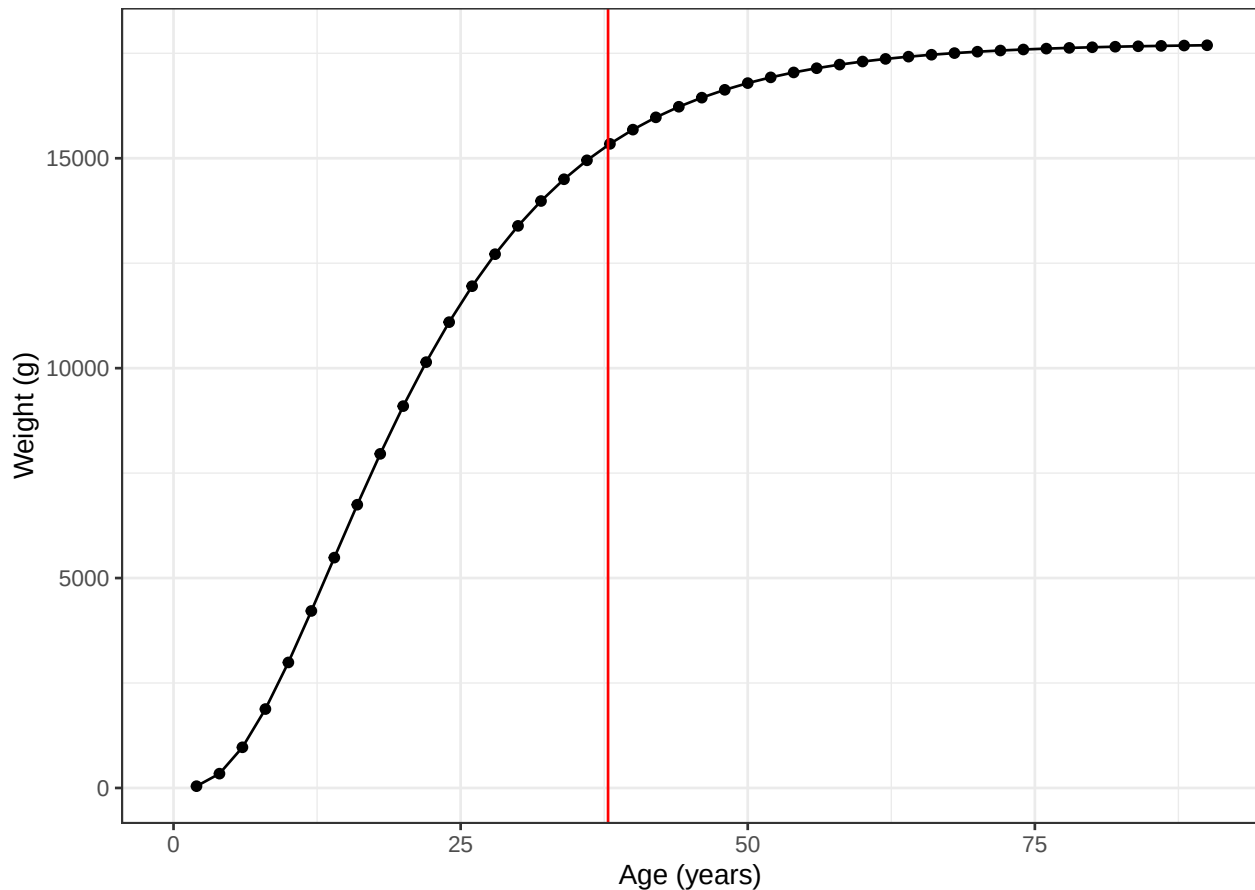
```
##           a      b
## 1 0.0259 2.858
data.frame(length = seq(0, lh$linf),
            weight = c(lh$a * seq(0, lh$linf)^lh$b)) %>%
  ggplot(aes(x = length, y = weight)) +
  geom_line() +
  labs(x = "Length (cm)", y = "Weight (g)") +
  theme_bw()
```



Weight at age

```
df_length$weight <- lh$a * df_length$length^lh$b

df_length %>%
  ggplot(aes(x = age, y = weight)) +
  geom_line() +
  geom_point(data = . %>% filter(age %in% agevec)) +
  geom_vline(xintercept = tmax, colour = "red") +
  labs(x = "Age (years)", y = "Weight (g)") +
  theme_bw()
```



Natural mortality

```
df_length$M <- rep(1.5 * lh["k"], nrow(df_length))

#df_length %>%
# ggplot(aes(x = age, y = M)) +
# geom_line() +
# geom_point(data = . %>% filter(age %in% agevec)) +
# geom_vline(xintercept = tmax, colour = "red") +
# labs(x = "Age (years)", y = "Natural mortality") +
# theme_bw() +
# coord_cartesian(ylim = c(0, 3))
```

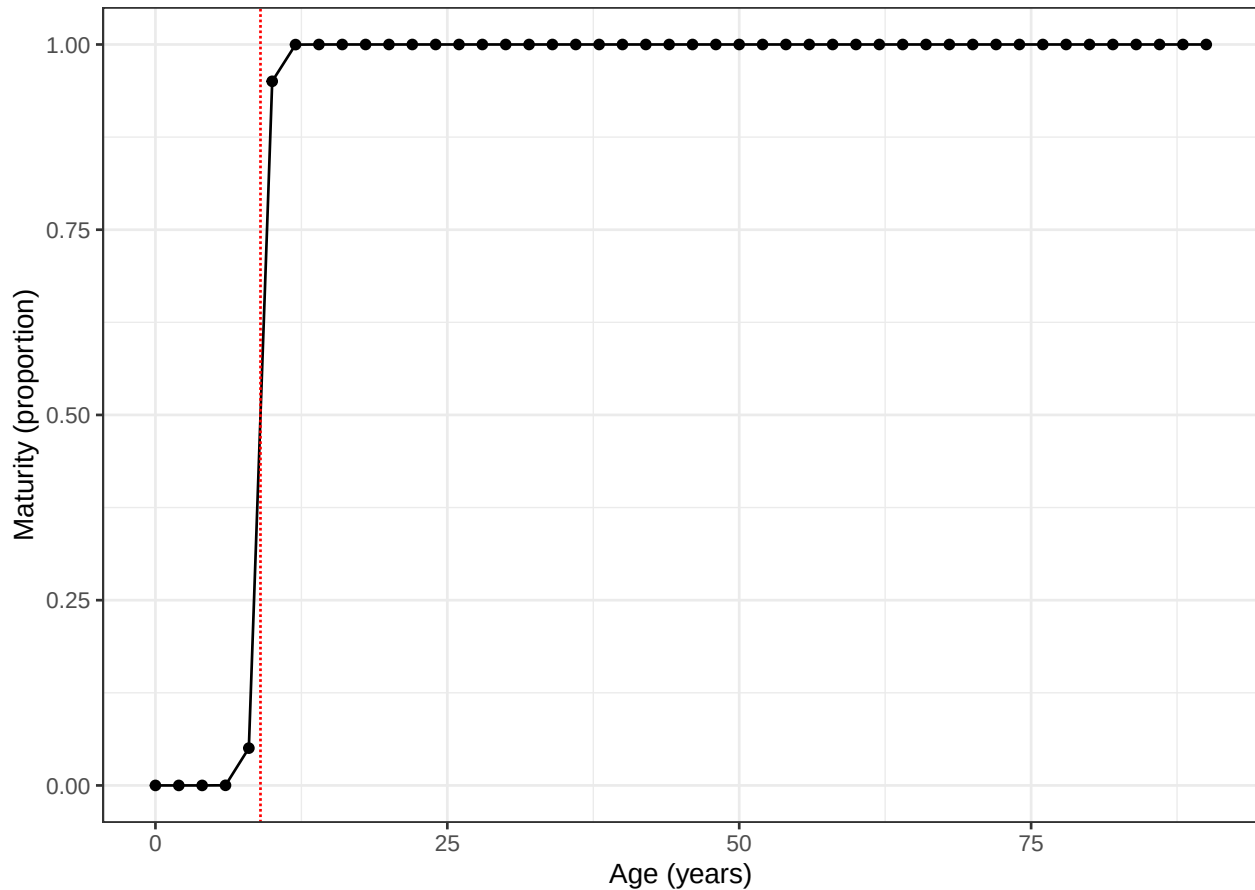
Maturity

```
# calculate a50
a50 <- vonB(len = FLQuant(lh$l50), params = FLPar(lh))

# create template for ages
ages_tmp <- FLQuant(agevec, dimnames = list(age = agevec))
mat_tmp <- FLlife::logisticFn(age = ages_tmp,
                             params = FLPar(a50 = a50, asym = 1, ato95 = 1))

as.data.frame(mat_tmp) %>%
```

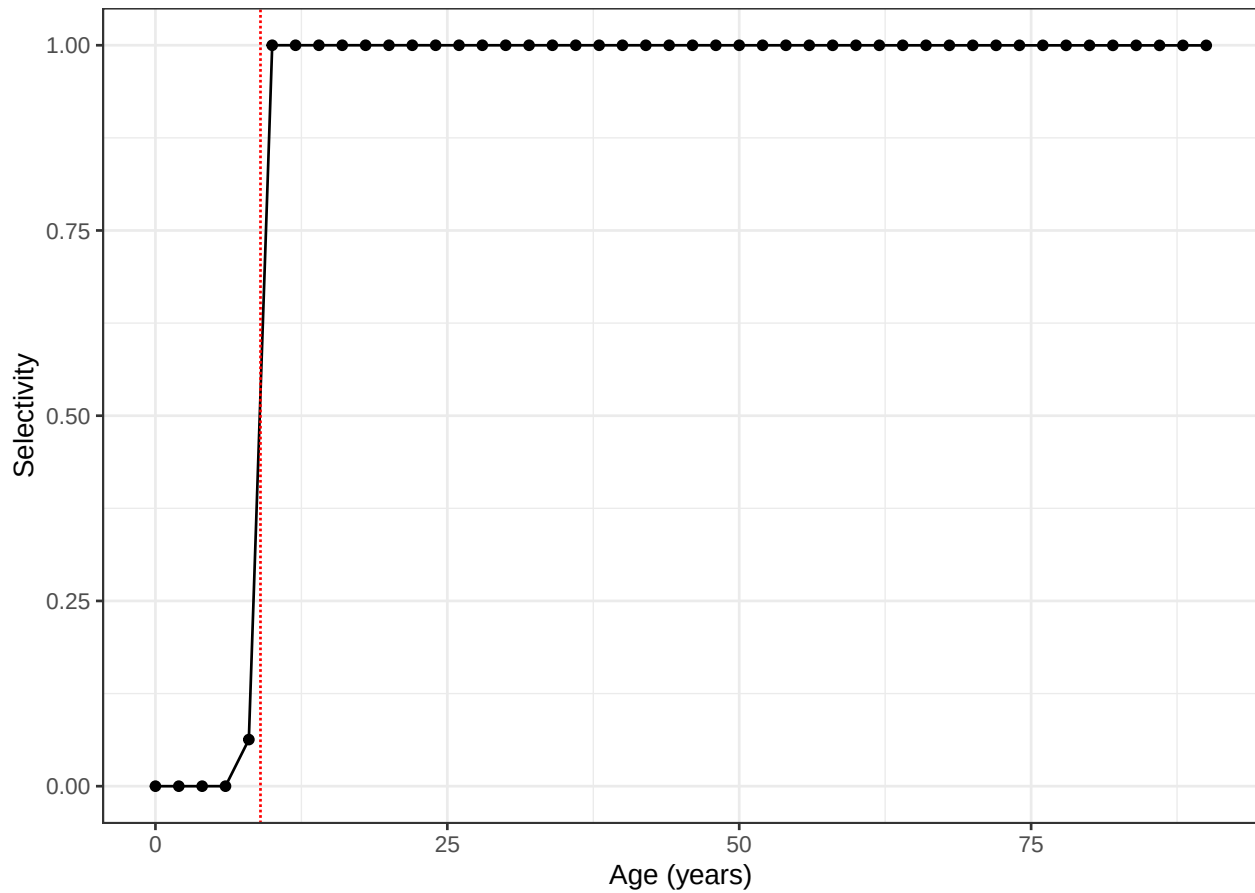
```
ggplot(aes(x = age, y = data)) +
  geom_line() +
  geom_point(data = . %>% filter(age %in% 0:90)) +
  geom_vline(xintercept = a50, colour = "red", linetype = "1111") +
  labs(x = "Age (years)", y = "Maturity (proportion)") +
  theme_bw()
```



Selectivity

```
# create template for ages
ages_tmp <- FLQuant(agevec, dimnames = list(age = agevec))
sel_tmp <- FLlife::dnormFn(age = ages_tmp,
  params = FLPar(sel1 = a50 + 1, sel2 = 1, sel3 = 5000))

as.data.frame(sel_tmp) %>%
  ggplot(aes(x = age, y = data)) +
  geom_line() +
  geom_point(data = . %>% filter(age %in% agevec)) +
  geom_vline(xintercept = a50, colour = "red", linetype = "1111") +
  labs(x = "Age (years)", y = "Selectivity") +
  theme_bw()
```

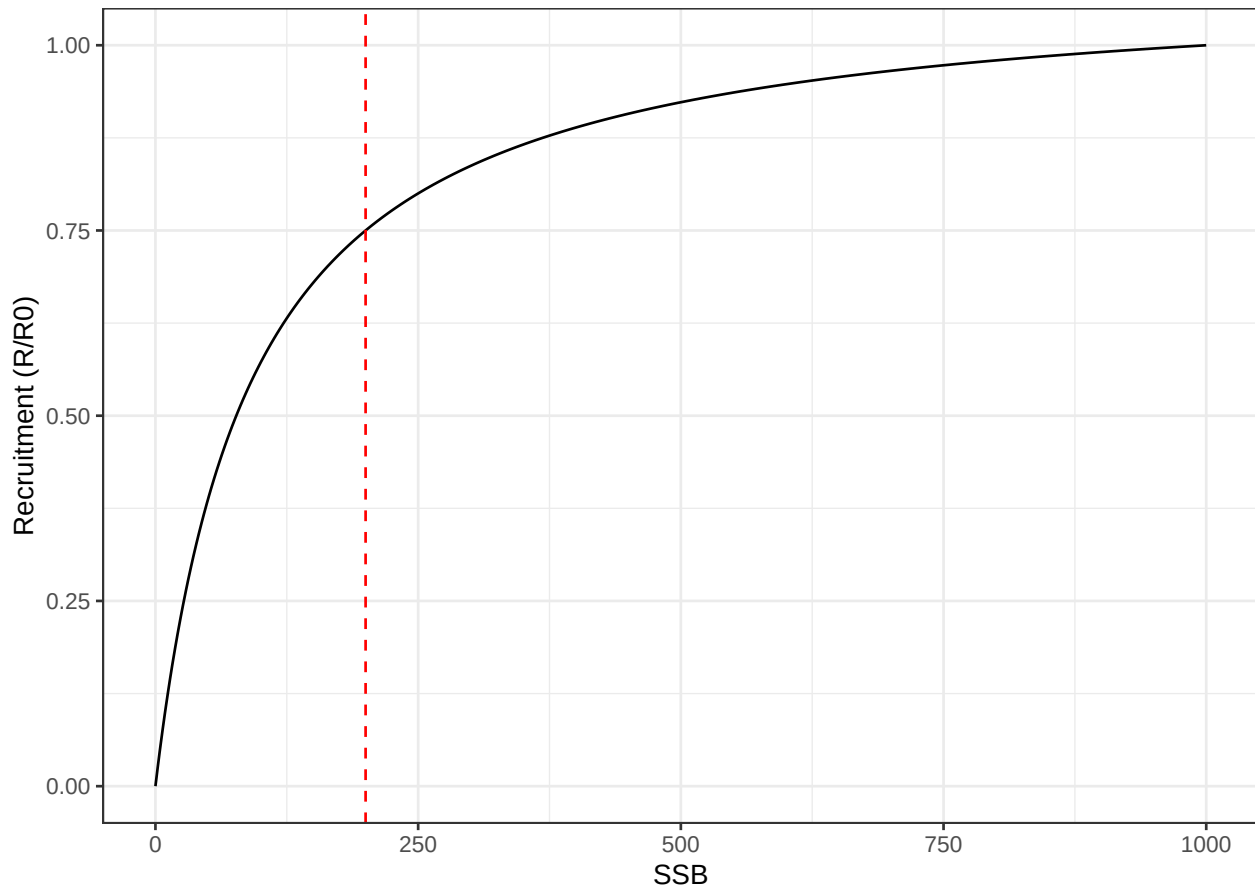


Stock-recruitment relationship

```
# define unfished biomass and recruitment steepness
lh$v <- 1000 ### unfished SSB (arbitrary)
lh$s <- 0.75 ### recruitment steepness (arbitrary)

# plot stock-recruitment relationship
# (relative to unfished recruitment, R0)
R0 <- 1 ### for plotting only
df_r <- data.frame(SSB = 0:1000)
df_r$R <- (0.8 * R0 * lh$s * df_r$SSB) /
  (0.2 * lh$v * (1 - lh$s) + (lh$s - 0.2) * df_r$SSB)

df_r %>%
  ggplot(aes(x = SSB, y = R)) +
  geom_line() +
  geom_vline(xintercept = 0.2 * lh$v, colour = "red", linetype = "dashed") +
  labs(x = "SSB", y = "Recruitment (R/R0)") +
  theme_bw()
```



Building the FLBRP and FLStock objects

```
# generate FLPar object with life-history parameters
lh_pars <- lhPar(lh[c("linf", "k", "t0", "a", "b", "150", "s", "v")])

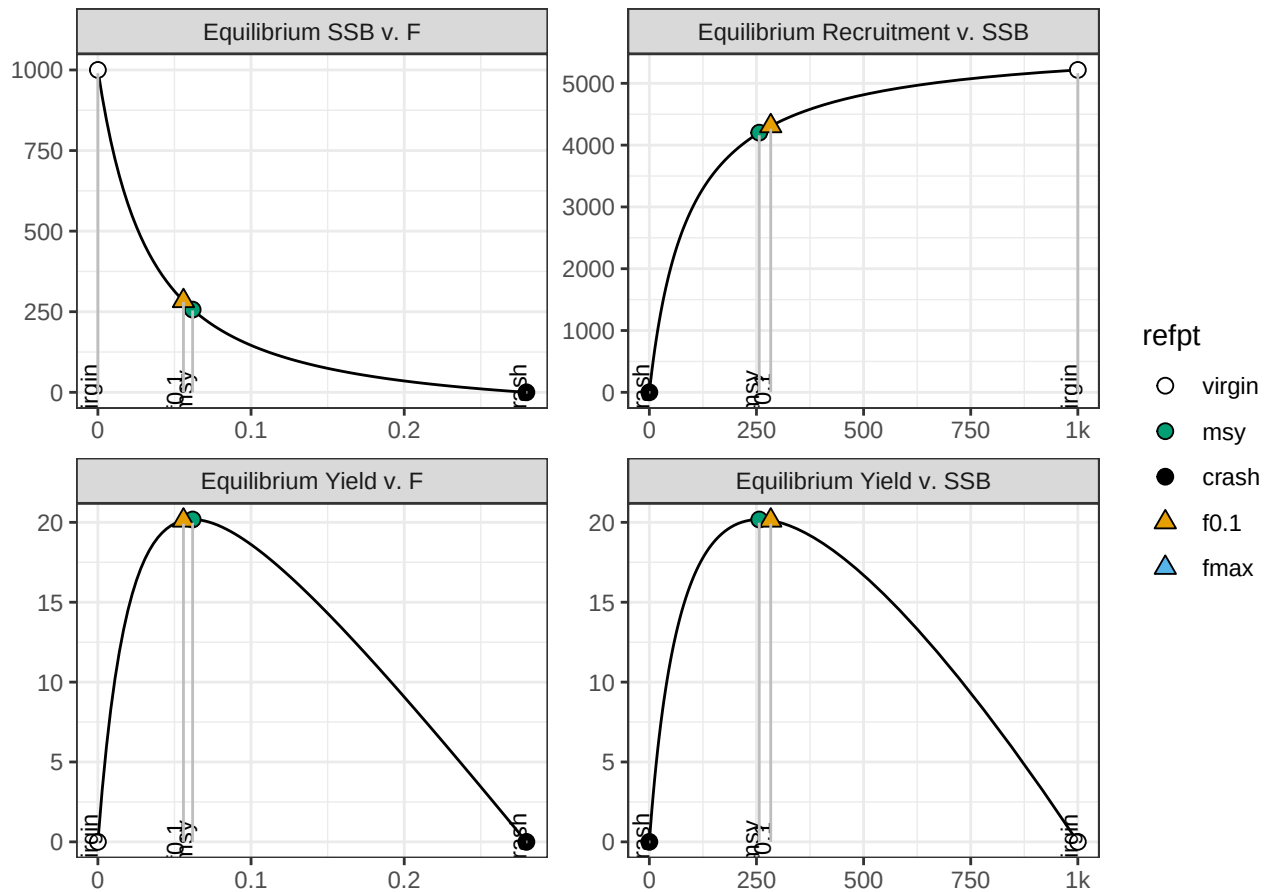
# create FLBRP with reference points
brp <- lhEq1(lh_pars, range = c(min = 1, max = ceiling(tmax),
                                minfbar = 1, maxfbar = ceiling(tmax),
                                plusgroup = floor(tmax)))

# show reference points
refpts(brp)
```

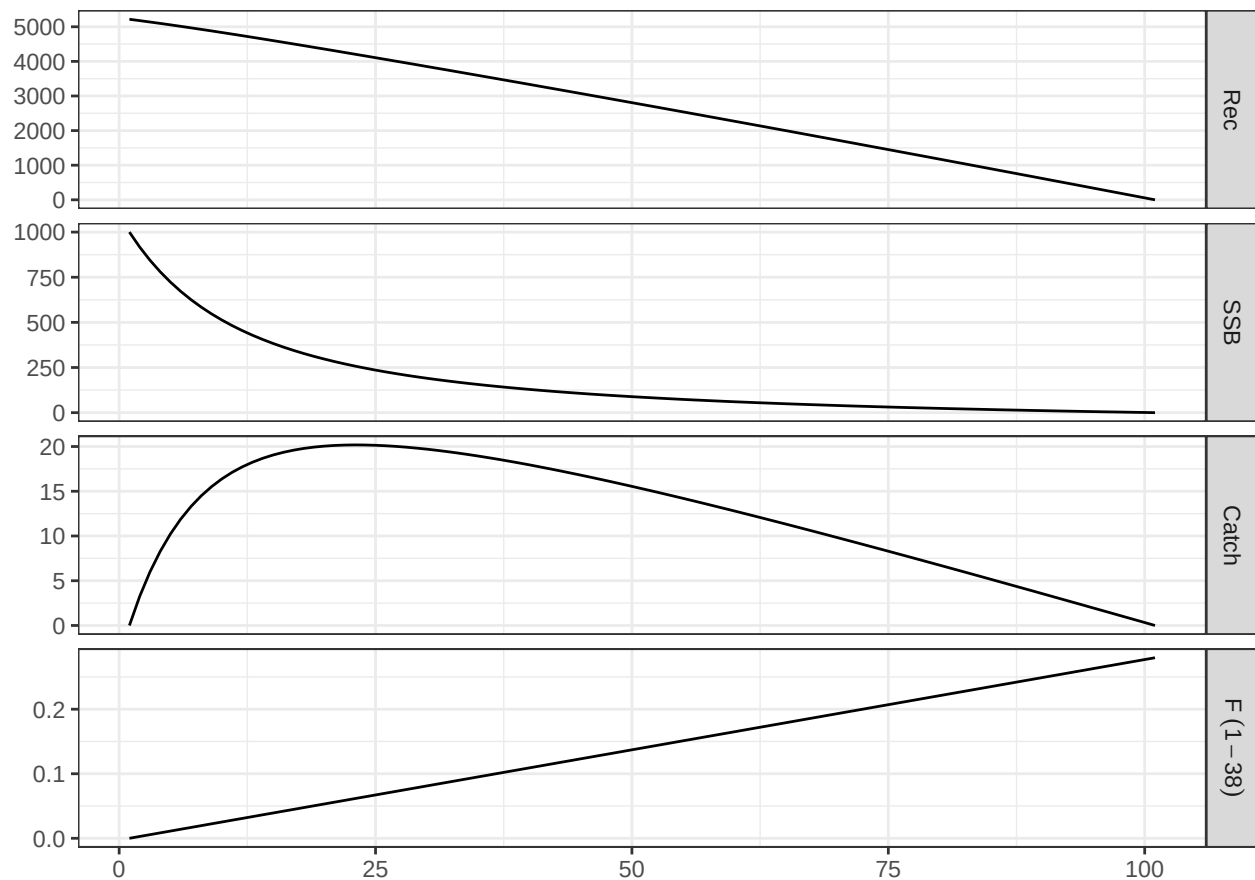
```
## An object of class "FLPar"
##      quant
## refpt  harvest  yield    rec    ssb    biomass  revenue  cost
## virgin  0.00e+00  0.00e+00  5.22e+03  1.00e+03  1.66e+04      NA    NA
## msy     6.18e-02  2.02e+01  4.20e+03  2.56e+02  1.28e+04      NA    NA
## crash   2.80e-01  1.72e-06  2.97e-04  4.74e-06  8.94e-04      NA    NA
## f0.1    5.59e-02  2.01e+01  4.31e+03  2.83e+02  1.32e+04      NA    NA
## fmax    5.00e-01 -2.63e+01 -4.52e+03 -4.03e+01 -1.36e+04      NA    NA
##      quant
## refpt  profit
## virgin      NA
## msy         NA
```

```
## crash      NA
## f0.1       NA
## fmax       NA
## units:    NA

# for plotting, remove Fmax
refpts(brp)["fmax", ] <- NA
plot(brp) + theme_bw()
```



```
# get stock
stk0 <- as(brp, "FLStock")
plot(stk0) + theme_bw()
```

```
# get stock-recruitment model object
sr <- FLSR(params = params(brp), model = model(brp))
model(sr)
```

```
## rec ~ a * ssb/(b + ssb)
## <environment: 0x2917daa8>
```

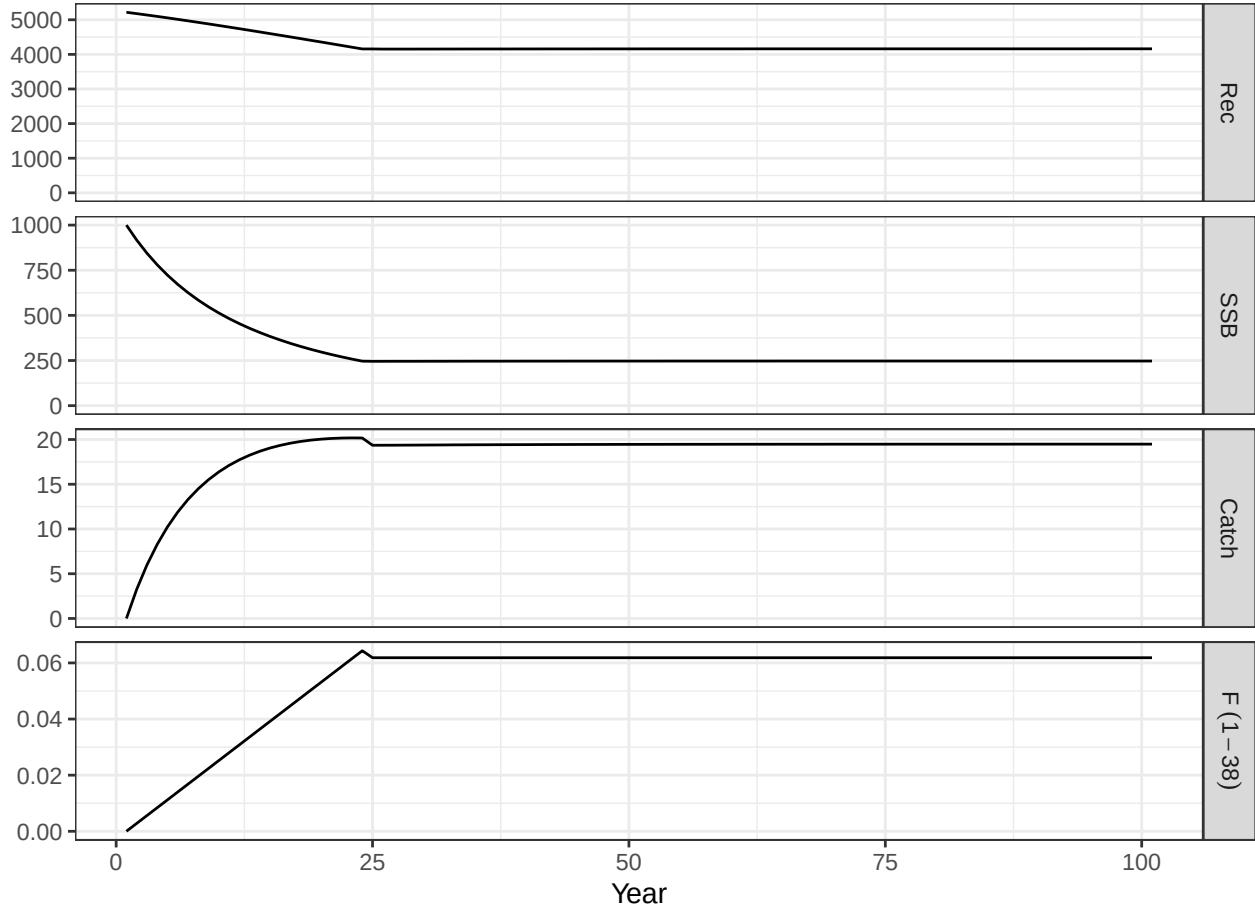
```
params(sr)
```

```
## An object of class "FLPar"
## params
##      a      b
## 5690.7  90.9
## units:  NA
```

Fishing history at Fmsy

```
# constant f at Fmsy
fmsy <- c(refpts(brp)["msy", "harvest"])

# apply fishing history to stock
ctrl <- fwdControl(year = 25:101, quant = "f", value = fmsy)
stk0 <- fwd(stk0, control = ctrl, sr = sr)
plot(stk0) + labs(x = "Year") + theme_bw()
```



Part 2: Deep-sea-like operating model

The OM has three areas: two submarine mounts subject to fishing (**a** and **b**) and one refuge area not subject to fishing (**c**). All three areas belong to the same stock (**sr** and **stk0** from Part 1).

Each scenario below varies the movement rates between areas via **args_om**. The OM setup is based on the following parameters:

- **smpf** — seamount partial f
- **bab** — movement rate from area a to area b
- **bba** — movement rate from area b to area a
- **bac** — movement rate from area a to area c
- **bbc** — movement rate from area b to area c
- **smpa** — seamount a partial R
- **smprb** — seamount b partial R
- **ftrga** — seamount a f target
- **ftrgb** — seamount b f target

Partial recruitment to area c is computed by $1 - \text{smpa} - \text{smprb}$.

There are three scenarios:

1. **Minimum movement**, with the same recruitment across areas. Areas a and b show the same behaviour. Biomass in area c is higher, since recruitment is the same across areas but area c is not fished.
2. **Strong vertical migration**, with the same recruitment across areas. Areas a and b become depleted, with very low biomass and catches reduced to very small amounts. Biomass in area c is very high, since

most of the population moves there and it is not fished.

3. **Strong horizontal movement.** Area a (donor) becomes depleted, with very low catches. In this case even recruitment is reduced, since horizontal migration includes recruitment and occurs after the settlement process. Area b (receiver) is in a healthy state, including high recruitment inherited from area a. Area c behaves as in scenario 01.

Scenario 01: minimum movement (by age and time)

```
args_om <- list(
  smpf = 0.5,      # seamount partial f
  bab = 0.05,      # from a to b
  bba = 0.05,      # from b to a
  bac = 0,         # from a to c
  bbc = 0,         # from b to c
  smpra = 0.333,   # seamount a partial R
  smprb = 0.333,   # seamount b partial R
  ftrga = 0.05,    # seamount a f target
  ftrgb = 0.05     # seamount b f target
)

# objects
stka_s01 <- stkb_s01 <- stkc_s01 <- stk_s01 <- stk0
sra.gm <- srb.gm <- src.gm <- as.FLSR(stk_s01, model = "geomean")
harvest(stka_s01) <- harvest(stk_s01) * args_om$smpf
harvest(stkb_s01) <- harvest(stk_s01) * (1 - args_om$smpf)
harvest(stkc_s01)[] <- 0

flq0 <- stock.n(stk_s01)
flq0[] <- range(stk_s01)["min"]:range(stk_s01)["max"]
# from a to b
bab <- flq0
bab[] <- args_om$bab
# from b to a
bba <- flq0
bba[] <- args_om$bba
# from a to c
bac <- FLife::logisticFn(age = flq0, params = FLPar(a50 = a50, asym = 1, ato95 = 1)) * args_om$bac
# from b to c
bbc <- FLife::logisticFn(age = flq0, params = FLPar(a50 = a50, asym = 1, ato95 = 1)) * args_om$bbc
mov <- FLQuants(bab = bab, bba = bba, bac = bac, bbc = bbc)

for (i in 50:100) {
  # common recruitment
  rfut <- predict(sr, ssb = ssb(stk_s01)[, ac(i - 1)])

  # movement
  # from a to b
  stknab <- stock.n(stka_s01)[, ac(i)] * mov[["bab"]][, ac(i)]
  # from b to a
  stknba <- stock.n(stkb_s01)[, ac(i)] * mov[["bba"]][, ac(i)]
  # from a to c
  stknac <- stock.n(stka_s01)[, ac(i)] * mov[["bac"]][, ac(i)]
  # from b to c
  stknbc <- stock.n(stkb_s01)[, ac(i)] * mov[["bbc"]][, ac(i)]
}
```

```

stock.n(stka_s01[, ac(i)] <- stock.n(stka_s01[, ac(i)] - stknab - stknac + stknba
stock.n(stkb_s01[, ac(i)] <- stock.n(stkb_s01[, ac(i)] - stknba - stknbc + stknab
stock.n(stkc_s01[, ac(i)] <- stock.n(stkc_s01[, ac(i)] + stknac + stknbc

# distribute recruitment evenly
params(sra.gm)[] <- rfut * args_om$smpra
params(srb.gm)[] <- rfut * args_om$smprb
params(src.gm)[] <- rfut * (1 - args_om$smpra - args_om$smprb)

# set projection at F level of ftrg in the seamounts, area c has
# harvest = 0, so no F will happen
ctrl <- fwdControl(year = c(i + 1), quant = "f", value = args_om$ftrga)
stka_s01 <- fwd(stka_s01, control = ctrl, sr = sra.gm)
ctrl <- fwdControl(year = c(i + 1), quant = "f", value = args_om$ftrgb)
stkb_s01 <- fwd(stkb_s01, control = ctrl, sr = srb.gm)
ctrl <- fwdControl(year = c(i + 1), quant = "f", value = 0)
stkc_s01 <- fwd(stkc_s01, control = ctrl, sr = src.gm)

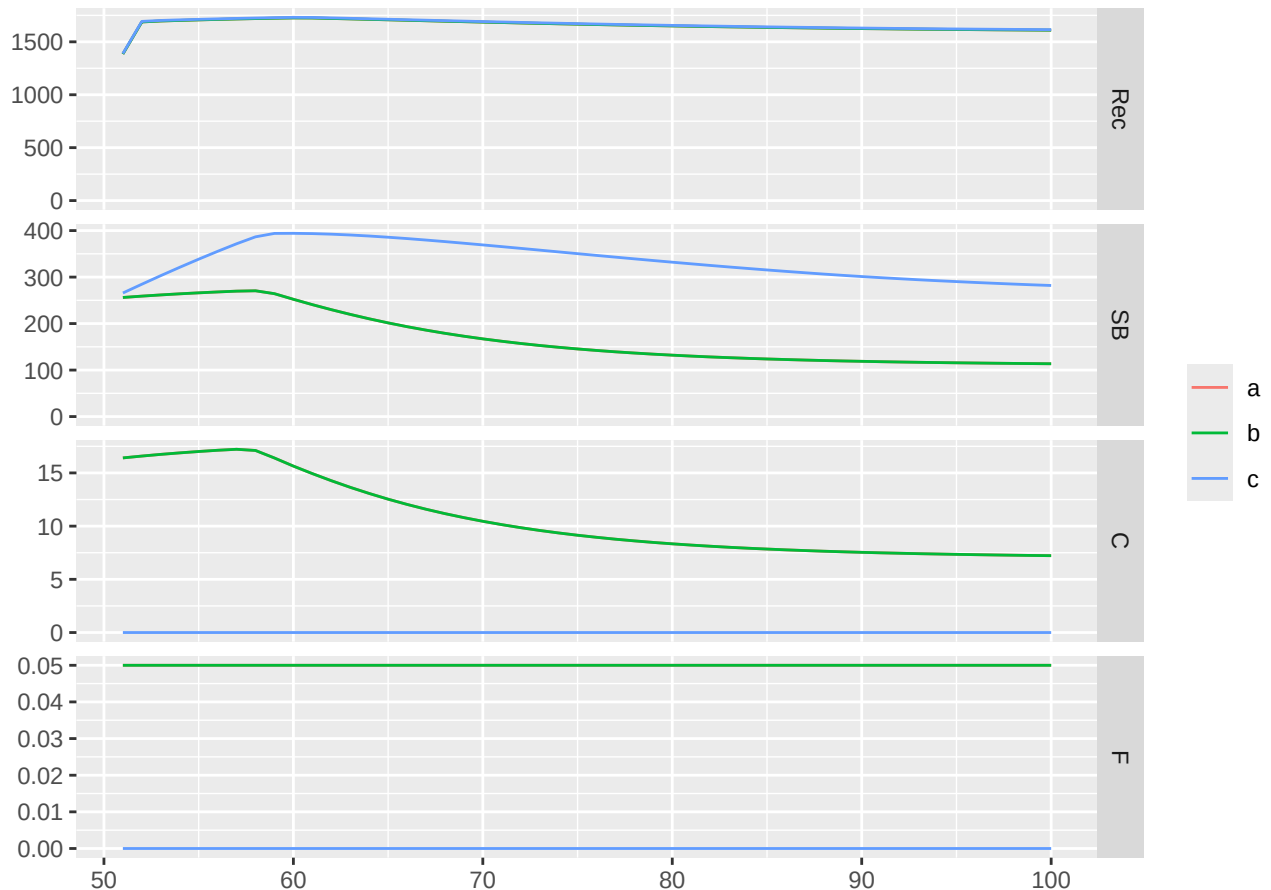
# update stock
stk_s01 <- stka_s01 + stkb_s01 + stkc_s01
}

```

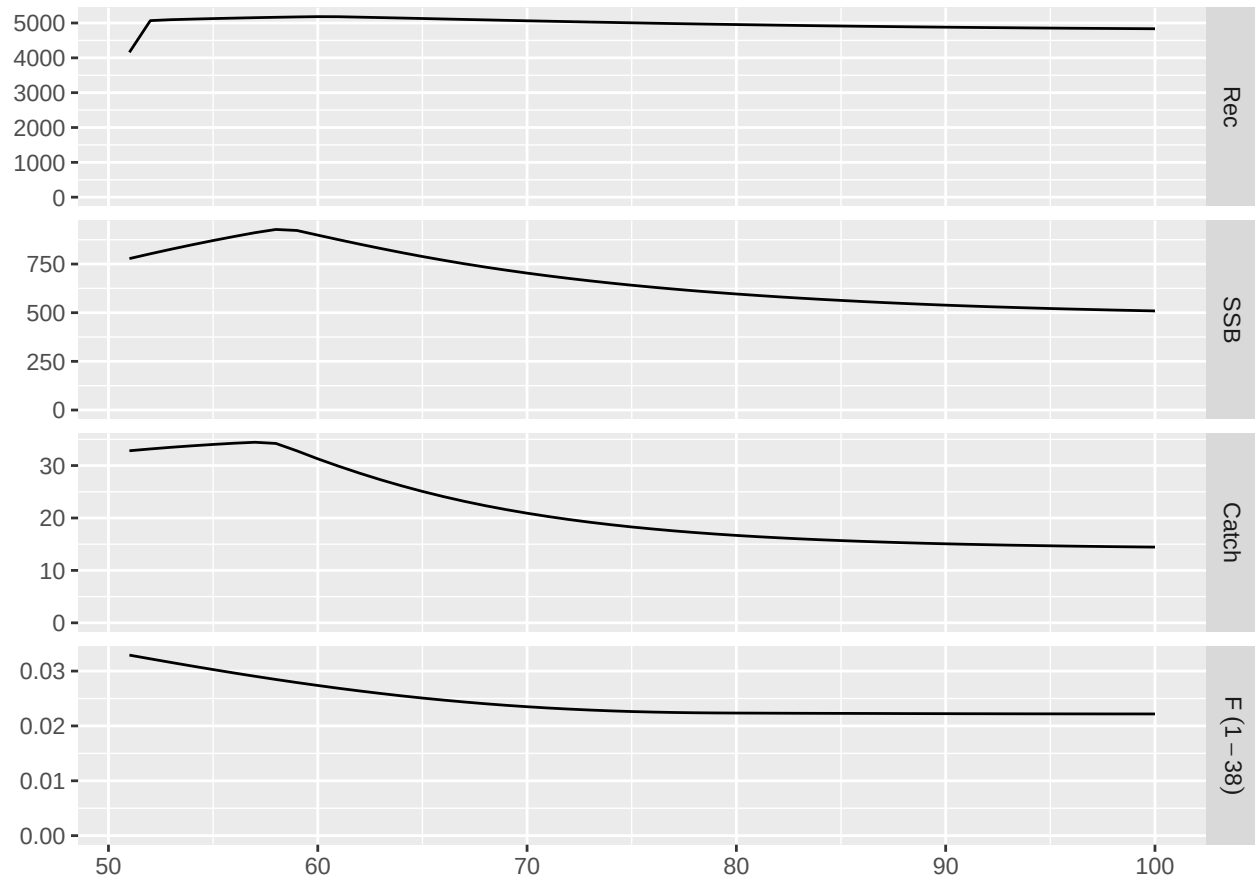
```

plot(window(FLStocks(a = stka_s01, b = stkb_s01, c = stkc_s01), 51, 100))

```



```
plot(window(stk_s01, 51, 100))
```



Scenario 02: strong vertical migration (by age and time)

```
args_om <- list(
  smpf = 0.5,      # seamount partial f
  bab = 0.05,     # from a to b
  bba = 0.05,     # from b to a
  bac = 1,        # from a to c
  bbc = 1,        # from b to c
  smpra = 0.333,  # seamount a partial R
  smprb = 0.333,  # seamount b partial R
  ftrga = 0.05,   # seamount a f target
  ftrgb = 0.05    # seamount b f target
)

# objects
stka_s02 <- stkb_s02 <- stkc_s02 <- stk_s02 <- stk0
sra.gm <- srb.gm <- src.gm <- as.FLSR(stk_s02, model = "geomean")
harvest(stka_s02) <- harvest(stk_s02) * args_om$smpf
harvest(stkb_s02) <- harvest(stk_s02) * (1 - args_om$smpf)
harvest(stkc_s02)[] <- 0

flq0 <- stock.n(stk_s02)
flq0[] <- range(stk_s02)["min"] : range(stk_s02)["max"]
# from a to b
```

```

bab <- flq0
bab[] <- args_om$bab
# from b to a
bba <- flq0
bba[] <- args_om$bba
# from a to c
bac <- FLife:::logisticFn(age = flq0, params = FLPar(a50 = a50, asym = 1, ato95 = 1)) * args_om$bac
# from b to c
bbc <- FLife:::logisticFn(age = flq0, params = FLPar(a50 = a50, asym = 1, ato95 = 1)) * args_om$bbc
mov <- FLQuants(bab = bab, bba = bba, bac = bac, bbc = bbc)

for (i in 50:100) {
  # common recruitment
  rfut <- predict(sr, ssb = ssb(stk_s02)[, ac(i - 1)])

  # movement
  # from a to b
  stknab <- stock.n(stka_s02)[, ac(i)] * mov[["bab"]][, ac(i)]
  # from b to a
  stknba <- stock.n(stkb_s02)[, ac(i)] * mov[["bba"]][, ac(i)]
  # from a to c
  stknac <- stock.n(stka_s02)[, ac(i)] * mov[["bac"]][, ac(i)]
  # from b to c
  stknbc <- stock.n(stkb_s02)[, ac(i)] * mov[["bbc"]][, ac(i)]

  stock.n(stka_s02)[, ac(i)] <- stock.n(stka_s02)[, ac(i)] - stknab - stknac + stknba
  stock.n(stkb_s02)[, ac(i)] <- stock.n(stkb_s02)[, ac(i)] - stknba - stknbc + stknab
  stock.n(stkc_s02)[, ac(i)] <- stock.n(stkc_s02)[, ac(i)] + stknac + stknbc

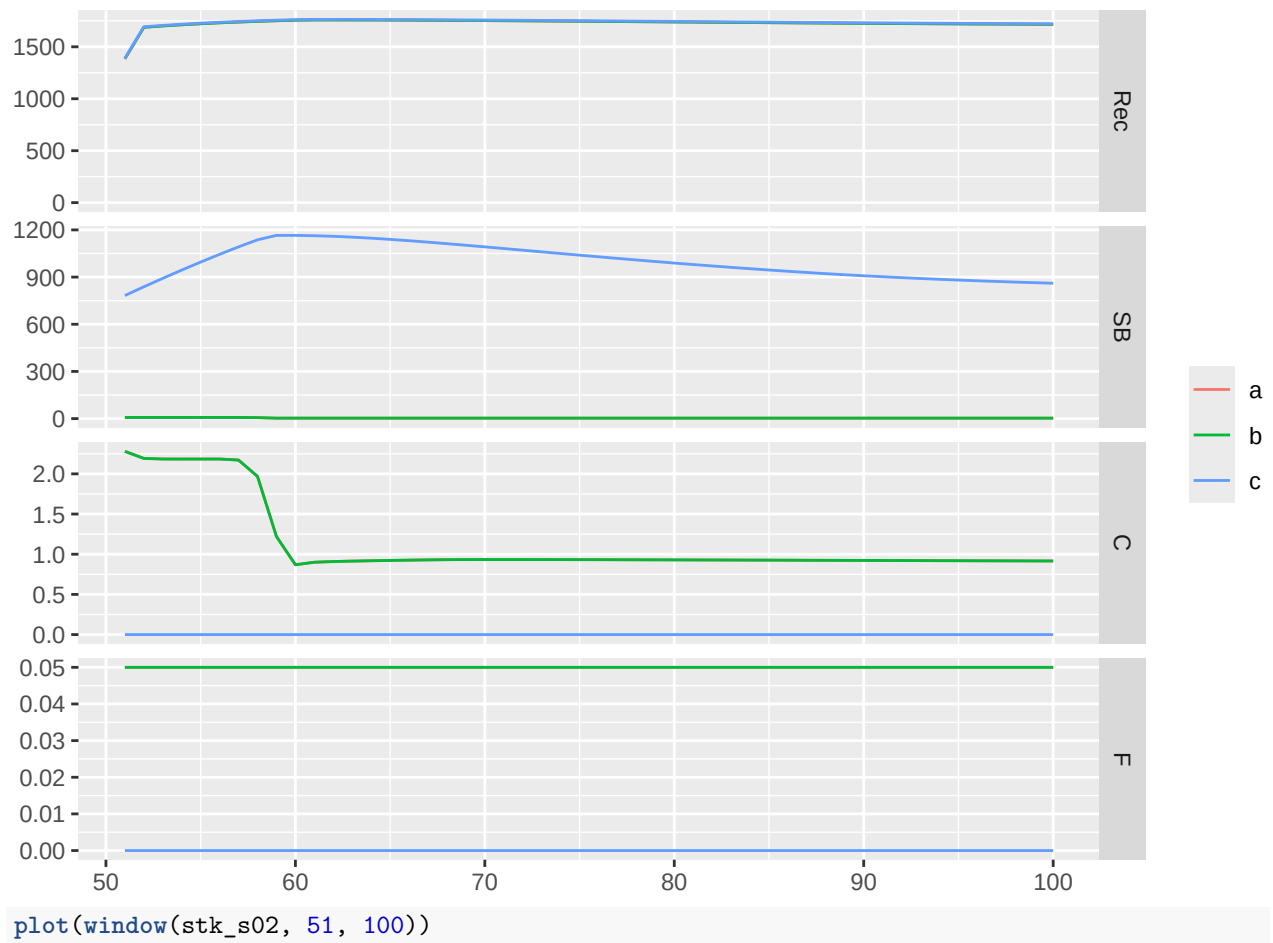
  # distribute recruitment evenly
  params(sra.gm)[] <- rfut * args_om$smpra
  params(srb.gm)[] <- rfut * args_om$smprb
  params(src.gm)[] <- rfut * (1 - args_om$smpra - args_om$smprb)

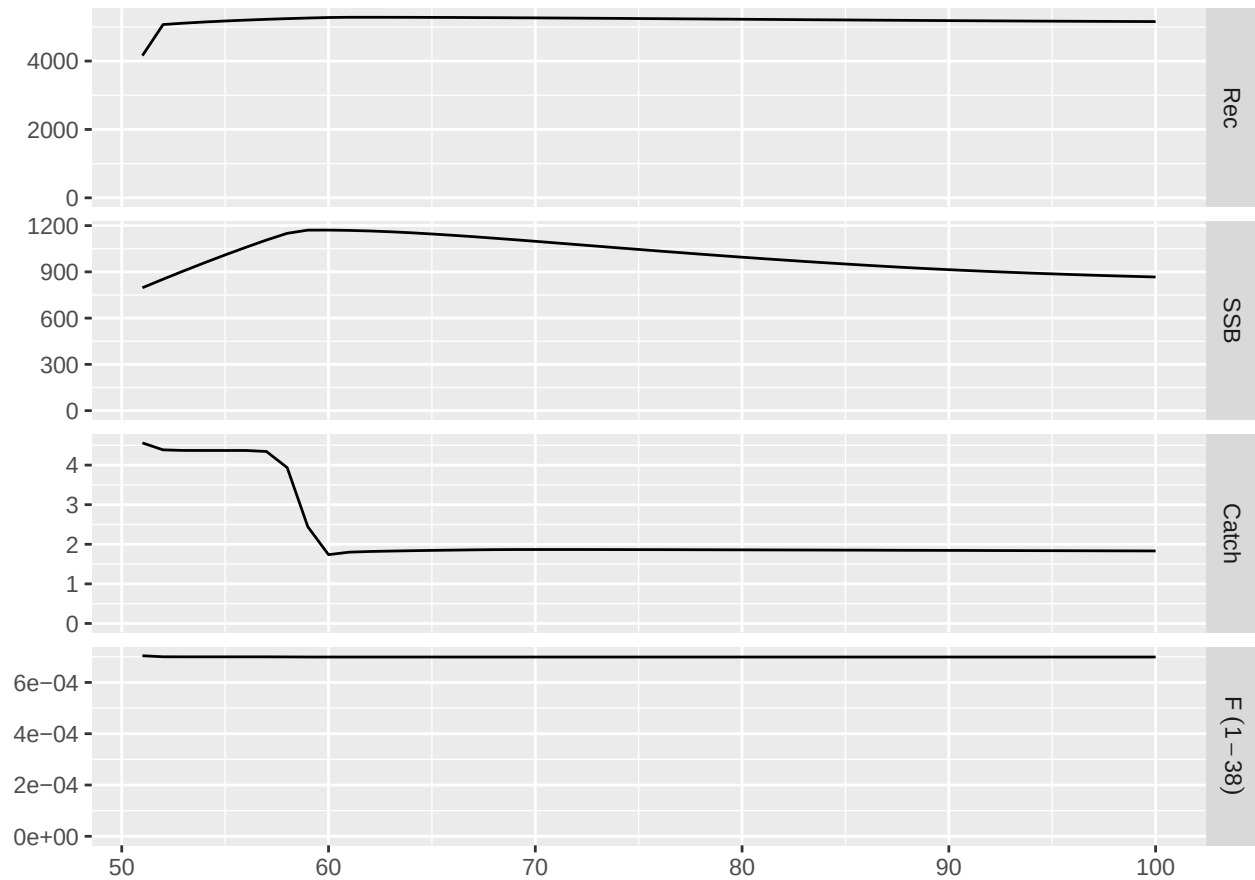
  # set projection at F level of ftrg in the seamounts, area c has
  # harvest = 0, so no F will happen
  ctrl <- fwdControl(year = c(i + 1), quant = "f", value = args_om$ftrga)
  stka_s02 <- fwd(stka_s02, control = ctrl, sr = sra.gm)
  ctrl <- fwdControl(year = c(i + 1), quant = "f", value = args_om$ftrgb)
  stkb_s02 <- fwd(stkb_s02, control = ctrl, sr = srb.gm)
  ctrl <- fwdControl(year = c(i + 1), quant = "f", value = 0)
  stkc_s02 <- fwd(stkc_s02, control = ctrl, sr = src.gm)

  # update stock
  stk_s02 <- stka_s02 + stkb_s02 + stkc_s02
}

plot(window(FLStocks(a = stka_s02, b = stkb_s02, c = stkc_s02), 51, 100))

```





Scenario 03: strong horizontal migration (by age and time)

```
args_om <- list(
  smpf = 0.5,      # seamount partial f
  bab = 0.75,      # from a to b
  bba = 0.05,      # from b to a
  bac = 0,         # from a to c
  bbc = 0,         # from b to c
  smpra = 0.333,   # seamount a partial R
  smprb = 0.333,   # seamount b partial R
  ftrga = 0.05,    # seamount a f target
  ftrgb = 0.05     # seamount b f target
)

# objects
stka_s03 <- stkb_s03 <- stkc_s03 <- stk_s03 <- stk0
sra.gm <- srb.gm <- src.gm <- as.FLSR(stk_s03, model = "geomean")
harvest(stka_s03) <- harvest(stk_s03) * args_om$smpf
harvest(stkb_s03) <- harvest(stk_s03) * (1 - args_om$smpf)
harvest(stkc_s03)[] <- 0

flq0 <- stock.n(stk_s03)
flq0[] <- range(stk_s03)[ "min" ] : range(stk_s03)[ "max" ]
# from a to b
bab <- flq0
bab[] <- args_om$bab
```



```

# from b to a
bba <- flq0
bba[] <- args_om$bba
# from a to c
bac <- FLife:::logisticFn(age = flq0, params = FLPar(a50 = a50, asym = 1, ato95 = 1)) * args_om$bac
# from b to c
bbc <- FLife:::logisticFn(age = flq0, params = FLPar(a50 = a50, asym = 1, ato95 = 1)) * args_om$bbc
mov <- FLQuants(bab = bab, bba = bba, bac = bac, bbc = bbc)

for (i in 50:100) {
  # common recruitment
  rfut <- predict(sr, ssb = ssb(stk_s03)[, ac(i - 1)])

  # movement
  # from a to b
  stknab <- stock.n(stka_s03)[, ac(i)] * mov[["bab"]][, ac(i)]
  # from b to a
  stknba <- stock.n(stkb_s03)[, ac(i)] * mov[["bba"]][, ac(i)]
  # from a to c
  stknac <- stock.n(stka_s03)[, ac(i)] * mov[["bac"]][, ac(i)]
  # from b to c
  stknbc <- stock.n(stkb_s03)[, ac(i)] * mov[["bbc"]][, ac(i)]

  stock.n(stka_s03)[, ac(i)] <- stock.n(stka_s03)[, ac(i)] - stknab - stknac + stknba
  stock.n(stkb_s03)[, ac(i)] <- stock.n(stkb_s03)[, ac(i)] - stknba - stknbc + stknab
  stock.n(stkc_s03)[, ac(i)] <- stock.n(stkc_s03)[, ac(i)] + stknac + stknbc

  # distribute recruitment evenly
  params(sra.gm)[] <- rfut * args_om$smpra
  params(srb.gm)[] <- rfut * args_om$smprb
  params(src.gm)[] <- rfut * (1 - args_om$smpra - args_om$smprb)

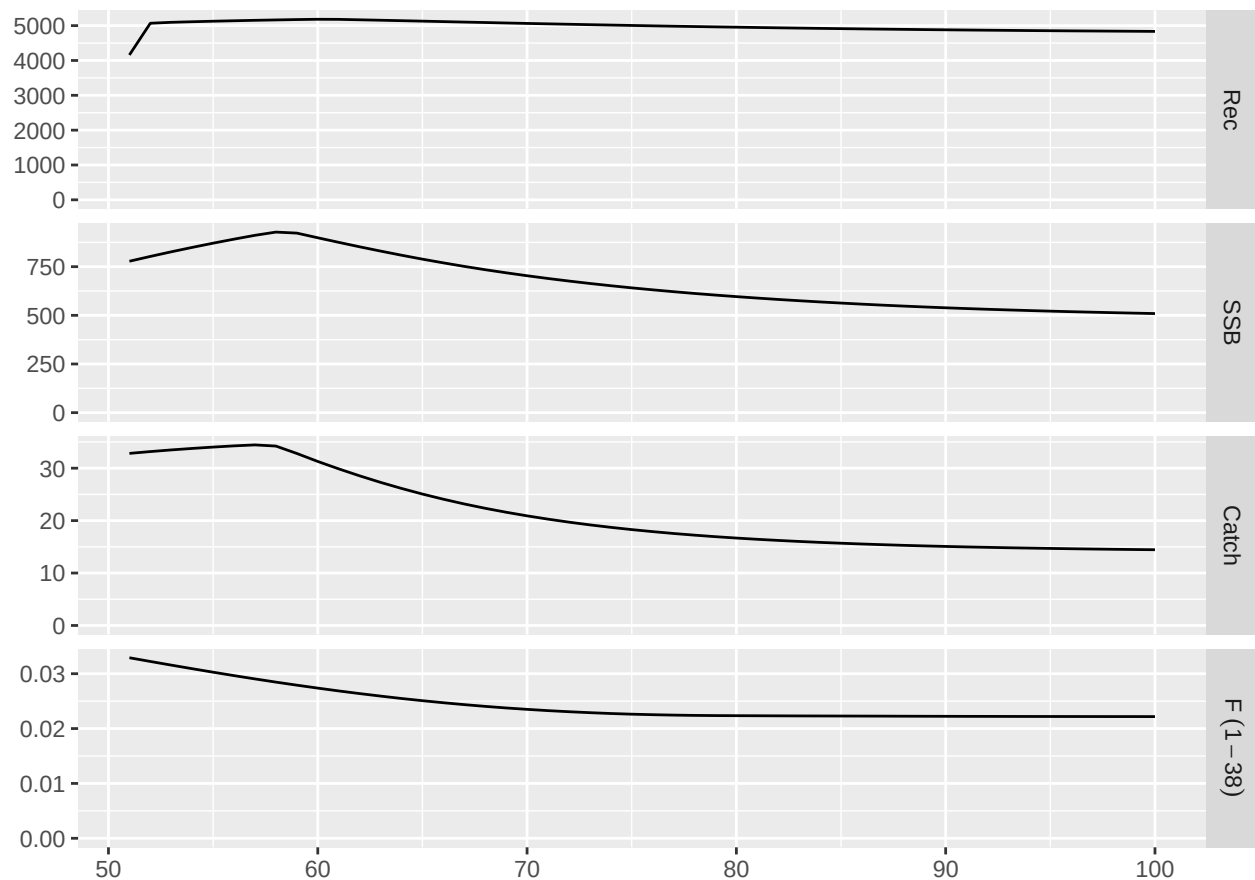
  # set projection at F level of ftrg in the seamounts, area c has
  # harvest = 0, so no F will happen
  ctrl <- fwdControl(year = c(i + 1), quant = "f", value = args_om$ftrga)
  stka_s03 <- fwd(stka_s03, control = ctrl, sr = sra.gm)
  ctrl <- fwdControl(year = c(i + 1), quant = "f", value = args_om$ftrgb)
  stkb_s03 <- fwd(stkb_s03, control = ctrl, sr = srb.gm)
  ctrl <- fwdControl(year = c(i + 1), quant = "f", value = 0)
  stkc_s03 <- fwd(stkc_s03, control = ctrl, sr = src.gm)

  # update stock
  stk_s03 <- stka_s03 + stkb_s03 + stkc_s03
}

plot(window(FLStocks(a = stka_s03, b = stkb_s03, c = stkc_s03), 51, 100))

```





Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: openSUSE Leap 16.0
##
## Matrix products: default
## BLAS: /usr/local/R452/lib64/R/lib/libRblas.so
## LAPACK: /usr/local/R452/lib64/R/lib/libRlapack.so; LAPACK version 3.12.1
##
## locale:
##  [1] LC_CTYPE=en_GB.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=en_GB.UTF-8      LC_COLLATE=en_GB.UTF-8
##  [5] LC_MONETARY=en_GB.UTF-8  LC_MESSAGES=en_GB.UTF-8
##  [7] LC_PAPER=en_GB.UTF-8     LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_GB.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Lisbon
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
```

```

##
## other attached packages:
## [1] dplyr_1.2.0      tidyr_1.3.2      FFlasher_0.7.3   FLFishery_0.4.0
## [5] FLife_3.4.0      ggplotFL_2.7.2   patchwork_1.3.2  ggrepel_0.9.8
## [9] ggplot2_4.0.3    FLCore_2.6.33    lattice_0.22-7   rmarkdown_2.30
##
## loaded via a namespace (and not attached):
## [1] sass_0.4.10      generics_0.1.4    digest_0.6.39     magrittr_2.0.5
## [5] evaluate_1.0.5    grid_4.5.2        RColorBrewer_1.1-3 iterators_1.0.14
## [9] fastmap_1.2.0     jsonlite_2.0.0    Matrix_1.7-4      tinytex_0.58
## [13] FLBRP_2.5.10      gridExtra_2.3     purrr_1.2.1       scales_1.4.0
## [17] jquerylib_0.1.4   cli_3.6.6         rlang_1.2.0       cowplot_1.2.0
## [21] withr_3.0.3       cachem_1.1.0      yaml_2.3.12       otel_0.2.0
## [25] tools_4.5.2       vctrs_0.7.3       R6_2.6.1          stats4_4.5.2
## [29] lifecycle_1.0.5   MASS_7.3-65       pkgconfig_2.0.3   pillar_1.11.1
## [33] bslib_0.10.0      gtable_0.3.6      glue_1.8.1        data.table_1.18.4
## [37] Rcpp_1.1.2        xfun_0.56         tibble_3.3.1      tidyselect_1.2.1
## [41] knitr_1.51        farver_2.1.2      htmltools_0.5.9   labeling_0.4.3
## [45] compiler_4.5.2    S7_0.2.2

```